Lesson 12: nth Term Test for Divergence

Assessment Criteria & Rubrics

- **Application Criteria:** The test must be applied to the sequence of terms a_n, not the partial sums S_n.
- **Divergence Criterion:** If $\lim_{n\to\infty} a_n = c$ and $c \neq 0$ (or if the limit does not exist, e.g., oscillates), the series $\sum a_n$ **diverges.**
- **Inconclusive Criterion:** If $\lim_{n\to\infty} a_n = 0$, the test is **inconclusive**. It cannot prove convergence.
- **Error Assessment:** A common error is declaring convergence when $\lim_{n\to\infty} a_n = 0$. The test only works as a **divergence** test.
- **Formal Proof Requirement:** The response must state that because $\lim_{n\to\infty} a_n \neq 0$, the series diverges by the nth-term test.